1. Clinical Question & Core Hypothesis

Clinical Question

Why do some tumors with favorable immune architecture still fail to achieve better clinical outcomes?

Working Hypothesis

Among HER2-positive tumors, higher pre-treatment stromal TIL density predicts greater trastuzumab-based therapy response, and this association is strongest when tumor ERBB2 expression is high and T cells exhibit a relatively non-exhausted state as indexed by low PDCD1 expression. Concretely, ERBB2-high, TIL-high, PDCD1-low tumors will have the highest probability of pathological complete response (pCR) to neoadjuvant trastuzumab-containing regimens; ERBB2-high but TIL-low or PDCD1-high tumors will show attenuated or absent benefit. Functionally, these functional TILs may facilitate ADCC and pro-immunogenic signaling upon trastuzumab engagement, enabling more robust anti-tumor activity. This testable model yields a specific interaction: (ERBB2 expression) × (TIL density) × (PDCD1 level) predicting therapy_response.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	ERBB2, PDCD1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ERBB2: ERBB2 as the target of trastuzumab and PDCD1 as a molecular marker that indexes T-cell functional state, both present in the available dataset.

PDCD1: PDCD1 as a molecular marker that indexes T-cell functional state, both present in the available dataset.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTLA4	Supporting modifier from knowledge graph
Secondary	LAG3	Supporting modifier from knowledge graph

Secondary	TIGIT	Supporting modifier from knowledge graph
Secondary	ESR1	Supporting modifier from knowledge graph
Secondary	COL1A1	Supporting modifier from knowledge graph
Secondary	FN1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: expression reflects T-cell functional state; higher PD-1 often indicates T-cell exhaustion, potentially limiting anti-tumor immunity. Mechanistically, abundant functional TILs adjacent to ERBB2-expressing cells could enhance antibody-dependent cytotoxicity and cytokine signaling, improving clinical response.

Prediction 2: may modulate immune accessibility or exhaustion and can be evaluated to refine the model. Experimental approach:

Prediction 3: Retrospective analysis of pre-treatment tumors with quantified ERBB2 and PDCD1 and objective measures of TIL density;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CTLA4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- LAG3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.